

Triplet and D1/D2 Local Correction Procedure

Legacy code `corr_tripD1.v6.f` (S. Fartoukh, March 2012)

1 Scope and Philosophy

The program `corr_tripD1.v6.f` computes local corrector settings for the HL-LHC insertion regions (IR1/2/5/8) to compensate multipole errors in the triplets and D1/D2. It follows the project note approach: for each IR, the left/right sides form a pair of correctors that cancel the dominant resonance driving terms (RDTs) coming from the corresponding magnet pair. The output is a MAD-X file `temp/MCX.setting.mad` with strengths for the dedicated corrector families (KQSX3, KCSSX3, KCOSX3, ...).

2 Indexing and Sides

- IR side index $i_c \in \{1, \dots, 8\}$: L1, R1, L2, R2, L5, R5, L8, R8.
- Magnet type index $im \in \{1, \dots, 6\}$: Q1, Q2a, Q2b, Q3, D1, D2.
- Slice index $s \in \{1, \dots, N_{\max}\}$ inside a magnet; $N_{\max} = 100$ in the code.
- Beam direction flag $b_v \in \{-1, 1\}$ detected from the order of elements; it flips the sign of skew/normal pairs when needed.

3 Optics Weights

From `temp/optics0_inser.mad` each slice provides (β_x, β_y, x, y) . Define

$$R_{ij} = \sqrt{\beta_x}^i \sqrt{\beta_y}^j = \beta_x^{i/2} \beta_y^{j/2}.$$

For feed-down terms a factor y multiplies the relevant weight (e.g. the dipole component of sextupoles). These weights are stored both for the main magnets (arrays `a?aux`, `b?aux`) and for the dedicated corrector families (arrays `a?c`, `b?c`). They represent the linear response of a given multipole to the chosen RDT component.

4 Error Aggregation

From `temp/tripD1D2.errors` each slice provides integrated normal (K_n) and skew ($K_n^{(\text{sk})}$) multipoles. For a given IR side i_c and multipole order n , the accumulated RDT contributions are

$$B_n(i_c) = \sum_s R_s^{(n)} K_n(s), \quad A_n(i_c) = \sum_s R_s^{(n)} K_n^{(\text{sk})}(s),$$

where $R_s^{(n)}$ is the appropriate optics weight for the resonance considered (see Table 1). Separate arrays are maintained for the left and right sides.

5 Pairwise Cancellation Strategy

For each IR, the left/right correctors form a two-element system that cancels two independent RDT components of the same multipole order. Let

- $c_L^{(1)}, c_R^{(1)}$ optics response of component 1 for left/right corrector,
- $c_L^{(2)}, c_R^{(2)}$ optics response of component 2 for left/right corrector,
- $b_L^{(1)}, b_R^{(1)}$ error contribution of component 1 (left/right),
- $b_L^{(2)}, b_R^{(2)}$ error contribution of component 2 (left/right).

Two patterns are used:

- **Difference cancellation** (antisymmetric): target $b_L^{(k)} - b_R^{(k)} = 0$ for $k = 1, 2$.
- **Sum cancellation** (symmetric): target $b_L^{(k)} + b_R^{(k)} = 0$ for $k = 1, 2$.

After choosing sum or difference, the required corrector strengths (k_L, k_R) solve the 2×2 system

$$\begin{bmatrix} c_L^{(1)} & c_R^{(1)} \\ c_L^{(2)} & c_R^{(2)} \end{bmatrix} \begin{bmatrix} k_L \\ k_R \end{bmatrix} = - \begin{bmatrix} b^{(1)} \\ b^{(2)} \end{bmatrix},$$

with determinant

$$\Delta = c_L^{(1)} c_R^{(2)} - c_R^{(1)} c_L^{(2)}.$$

The code implements

$$k_L = \frac{-c_R^{(2)} b^{(1)} + c_R^{(1)} b^{(2)}}{\Delta}, \quad k_R = \frac{-c_L^{(1)} b^{(2)} + c_L^{(2)} b^{(1)}}{\Delta},$$

where $b^{(k)}$ are either $(b_L^{(k)} - b_R^{(k)})$ or $(b_L^{(k)} + b_R^{(k)})$ depending on the symmetry. The beam-direction flag b_v flips the sign of some solutions so they follow the MAD-X convention for the two beams.

6 Targets and Correctors

Table 1 summarizes the components cancelled, the symmetry used, and the corrector families produced in `MCX_setting.mad`.

7 Outputs

- `temp/MCX_setting.mad`: MAD-X assignments (e.g. `KQSX3.L1 := value / 1.MQSXF;`) for every IR side, scaled by the element length.
- Console diagnostics: slice counts, beam direction, warnings on missing entries, and a detailed RDT dump before/after correction for all targeted components.

Multipole	Components cancelled	Corrector families (per IR side)
Skew quad $a2$	$(1, -1)$ local feed-down	KQSX3 (all IRs)
Skew sext $a3$	$(0, 3), (3, 0)$ difference	KCSSX3 (all IRs)
Skew oct $a4$	$(1, 3), (3, 1)$ sum	KCOSX3 (all IRs)
Skew deca $a5$	$(0, 5), (5, 0)$ difference	KCDSX3 (IR1/5)
Skew 14-pole $a6$	$(5, 1), (1, 5)$ sum	KCTSX3 (IR1/5)
Normal sext $b3$	$(1, 2), (2, 1)$ difference	KCSX3 (all IRs)
Normal oct $b4$	$(0, 4), (4, 0)$ sum	KCOX3 (all IRs)
Normal deca $b5$	$(5, 0), (0, 5)$ difference	KCDX3 (IR1/5)
Normal 14-pole $b6$	$(6, 0), (0, 6)$ sum	KCTX3 (all IRs)

Table 1: RDT components cancelled and associated corrector families. Orders follow the notation (i, j) for $x^i y^j$ terms in the multipole expansion; “difference” means left/right subtraction, “sum” means left/right addition.

8 Practical Notes

- Slice alignment: the optics table and error table must have matching slice counts for each magnet type; otherwise the program stops.
- Expected entry counts: 836 optics slices and 776 error entries in the reference setup; deviations trigger warnings.
- Feed-down: sextupole dipole feed-down uses the measured orbit y in the optics file; left/right feed-down terms are swapped to maintain symmetry.
- Beam direction: b_v is inferred from the sequence (incrementing or decrementing i_c) and is applied to preserve the sign conventions between the two beams.